Mathematics or the Verifier? A Null Result for One Verifier-Filtered Self-Training Recipe on a 1.5B Model

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Abstract. Self-training and reinforcement learning with verifiable rewards report large gains for small language models, but a rising score does not show that a model learned mathematics rather than what its verifier accepts. We study the cheapest form of the loop, generate–verify–select–train with offline LoRA SFT, on Qwen2.5-1.5B-Instruct with the rule-based verifier `math-verify`, on held-out MATH-500 and GSM8K exams (150 problems; 500 in a final round). No pass@1 or pass@k gain at any of four learning rates has a problem-level bootstrap interval excluding zero; in the final round, with a revised recipe, pass@1 falls 3.2–4.8 points in each of three seeds (mean -4.1). The design detects only gains of about 10 points of pass@1 (6 in the final round). A positive gate fails: training on reference solutions lowers pass@1 by 32.2 points. The boxed share is unchanged (90.6 $\rightarrow$ 90.3%) while accuracy among boxed answers falls (48.5 $\rightarrow$ 17.1%) and solutions shorten 3.2$\times$, so the model imitates terse reference solutions and loses reasoning, which makes the gate a weak control. The null therefore concerns this recipe only and cannot show that self-training fails. Scores are graded by the verifier that selects the data (the final round also by a string-match grader), so neither the mathematics nor the verifier question is answered. A post hoc audit of a gold-storage defect in our pipeline (about one in six correct solutions rejected) shows a fall tied by an identity to the falling `\boxed` share, and is uninformative about adaptation. Patching a rarer defect and removing the filter show no consistent difference but are dose-confounded or too small to detect one.

Keywords: self-training · self-improvement · verifier bias · mathematical reasoning · small language models · verifiable rewards · null result.

1 Introduction

A common recipe for improving the mathematical reasoning of a small language model (SLM) *without human annotation* is to let the model generate its own solutions, keep those an automated verifier judges correct, and fine-tune on them, for several rounds. We call this the *Generate–Verify–Train* (GVT) loop; it underlies STaR [19], ReST [12], ReST-EM [11] and rejection-sampling fine-tuning [17]. But the verifier never reads the reasoning: a rule-based verifier such as `math-verify`[3] only extracts a final answer and compares it with the gold, and it rejects some correct answers written in forms it does not parse (Section 3.2). The training signal is "was the final string accepted", not "were the steps right".

The same gap runs through reinforcement learning with verifiable rewards (RLVR): only 8.6 of one study's 37.6 points lie beyond format correction [14], and GRPO with *random* rewards raises Qwen2.5-Math-7B by 21.4 points on MATH-500, against 29.1 with ground-truth rewards [9]. A rising score does not by itself show that a model learned to solve more problems rather than to surface behaviour it had or to produce what its reward accepts. We study the question in a cheap and common form of the loop, offline rejection-sampling SFT, where a false reject removes a training example, and apply a per-round false-reject measurement there, post hoc, for one defect of our pipeline.

We ask: *after several GVT rounds, is the model better at mathematics, or has it only learned to write what the verifier accepts?* We fixed four measures in advance: (i) pass@1; (ii) pass@k; (iii) output style, chiefly the share of generations containing `\boxed`; and (iv) per-problem transitions (problems *added* and *lost* between rounds). The pre-set reading rule was that "learning the verifier's style" requires pass@1 to rise, pass@k to stay flat and style to converge at once. In hindsight the rule is one-sided: it can only fire when accuracy rises (Section 5.1). The higher learning rates, the controls, sample accuracy as primary measure, the termination measures, the re-scoring and the final 500-problem round (Section 4.6) were added after seeing earlier runs; all are post hoc.

We detect no improvement at any learning rate, though gains below about 10 points of pass@1 or 3.5–5.5 of sample accuracy could have been missed (intervals omit training-seed variance), and no arm of this recipe improves, so the result concerns the recipe. The second branch is untested: no run varies the frequent verifier defect, scores are graded by the selecting verifier (one string-match check on the final round), and the post hoc audit of Section 5.3 is uninformative about adaptation. Most clearly at the highest rate, accuracy falls on the solutions the model does finish, though on one seed the cap share carries more of it (Section 4.5). A sharp pooled decline is absent in the isolated runs at 10^{-5}, which train on about 4.5× fewer examples; dose and pooling are confounded.

Contributions.

[3] https://github.com/huggingface/math-verify, version 0.9.0 with antlr4 4.13.2 in this work.

- For a 1.5B model, LoRA SFT and one rule-based verifier, no gain is detectable on a held-out exam at any learning rate we tried (a coarse grid that also changes k and rounds), with bootstrap intervals and the smallest effects the design could detect. No arm improves, not even training on reference solutions, so this is a null for the recipe, not a statement about self-training.
- A post hoc per-round false-reject audit of one verifier defect (the RLVR tracking [4,16] applied to offline SFT; not a novel method). The rejected share falls by 1.5–2.6 points with intervals reaching zero, tied to the falling `\boxed` share by an identity; the audit is uninformative about adaptation (Section 5.3).
- A decomposition of an accuracy decline under a fixed token cap into a finished-accuracy part and a cap-share part: at the highest rate, finished generations lose 5.5–10.6 points against the untrained model on the same problem mix (A, B and no-filter runs; train-on-gold about 26).

2 Related Work

Self-training and its limits. STaR [19] introduced the generate–filter–retrain loop, and ReST [12], ReST-EM [11] and RFT [17] formalise it; ReST-EM, which like us restarts from the base model, reports diminishing gains after the first round. Yue et al. [18] show that RL raises pass@1 while the base model wins at large pass@k; Teacher-Free Self-Training [13] concludes that self-training "amplifies but does not compound". Repeated training on model-generated data erodes distribution tails [10], and a narrow selector can accelerate this [8].

RL with verifiable rewards at small scale. RLVR with GRPO reports large gains near our model size: one training example lifts Qwen2.5-Math-1.5B on MATH-500 from 36.0% to 73.6% [14], 7,000 examples lift DeepSeek-R1-Distill-Qwen-1.5B on AMC23 from 63% to 80% [3], and zero-RL works on open base models from 0.5B [20]. These studies improve accuracy, which our SFT loop does not, though of Wang et al.'s 37.6 points only 8.6 lie beyond format correction, and random rewards also help [9]. We do not compete with RLVR on accuracy.

Verifier error. TinyV [16] finds that rule-based verifiers reject many correct solutions; From Accuracy to Robustness [6] reports an average recall of 86%, and 0.95 for `math-verify` on their MATH subset. Imperfect Verifiers [1] models fixed false-accept and false-reject rates. Egashira et al. [4] find in RLVR that systematic false negatives behave like random noise whereas false positives can cause plateau or collapse, and that the policy first unlearns format-triggering behaviour. Our setting is offline SFT with a fresh adapter per round, and our controls cannot test their findings (Section 4.3).

Auditing self-improvement. Xu et al. [15] audit LoRA self-training of Qwen3-8B against a frozen control and identify seven measurement failures, including training-seed variance and a fixed token cap under style drift. This is the study

closest to ours; it is why we separate finished accuracy from termination (Section 4.5). We differ in scale (a 1.5B model) and in the verifier (a rule-based one whose rejection defect we measured). Kim et al. [7] show that self-distillation with a teacher that sees the solution can cut accuracy; our train-on-gold drop resembles this, but we did not test its cause. Our per-round false-reject measurement is not new (it is the per-step tracking of Egashira et al. and TinyV, applied to offline SFT and to a defect of our own pipeline); we claim neither it nor the patch control as a contribution. Supplementary Appendix A gives the full comparison.

3 Setup

3.1 The GVT Loop

Each round t has four steps. *Generate:* the current model samples k solutions per problem. *Verify:* `math-verify` extracts the final answer and compares it with the gold. *Select:* accepted solutions are kept. *Train:* a *fresh* LoRA adapter is attached to the *base* model and trained on the kept solutions for one epoch. Round 0 is the untrained base model; the round-t model ($t \geq 1$) is the base plus an adapter trained on the pool accepted at round $t-1$. Adapters are not accumulated, so earlier rounds affect later ones only through the data they generate; this restart-from-base design gives compounding less room than continued training would.

3.2 The Verifier and Its Directional Bias

A format probe that re-renders gold answers in valid forms shows that `\dfrac` and `\tfrac` are always rejected, on both datasets. On real solutions (one public Qwen2.5-7B-Instruct solution for each of 200 MATH-500 problems, judged correct by a normalised comparison) the false-negative rate is 7.8% (10 of 128; Wilson interval 4.3–13.8%) on MATH-500 and 0% on GSM8K. Eight of the ten fail when the whole solution is verified but pass when only the boxed answer is, probably the gold-storage defect described below. The audit inspected only rejections, so it gives no false-positive rate. The verifier also cannot parse 51 of the 500 stored MATH-500 golds (10.2%); under V these always count as unsolved and never enter the training pool (V' makes 46 parseable for boxed predictions). This cancels in within-run differences only if they would not otherwise change (unchecked; 11–18 per exam partition).

The patch. For the control we rewrote `\dfrac` and `\tfrac` to `\frac` before parsing, and nothing else.

The gold-storage defect (post hoc). Our pipeline stores the gold without a delimiter, and `math-verify` applies its full LaTeX extraction only inside `\boxed{}` or `$...$`, so a correct boxed prediction can fail against a bare gold (e.g. the prediction `\boxed{\frac{4}{3}}` against the stored gold `\frac43`). Re-scoring the saved generations with a verifier V' that wraps the gold in `\boxed{}` when the prediction contains one shows that this defect rejects 15.5–18.3% of the generations V' accepts. It would enlarge the round-0 kept pool by about a fifth to a

quarter (440 → 542, +23%; selection problems only, so its 18.8% rejected share differs from the 15.5–18.3% over all 500), against 0–4 examples for the `\dfrac` patch, and it lowers absolute pass@1 by 4.7–8.0 points in self-training. It is bound to the problem, not to the answer: on the 54–62 problems where it ever fires, it rejects 94–98% of correct answers in every round, and 29–34 of these problems have a gold the original verifier cannot parse at all, so no answer form avoids it there. Re-scored, sample accuracy still falls in all five runs we re-scored (three self-training, two train-on-gold); the original-verifier changes are 0.73–0.85 of the re-scored ones, a ratio we do not use to rescale other numbers. All tables keep the original verifier, the one the loop trained with. Supplementary Appendix B gives the full analysis.

3.3 Two Experimental Designs

Design P (pooled) loads both datasets (500 problems each) together and trains *one* adapter per round on the union of accepted solutions, with selection and evaluation on the same 1,000 problems. *Design I (isolated)* uses one dataset and one adapter per run, and holds out 30% of the 500 problems (150) as an `exam` set; generation runs on all 500 problems each round, but only the other 350 can enter the training pool, and all Design I pass rates are on the exam set. Table S1 summarises them.

Design I has three runs, all Qwen2.5-1.5B-Instruct, five rounds, $k = 8$, differing in one respect each: **Run A** (MATH-500, original verifier), **Run B** (MATH-500, patched verifier) and **Run C** (GSM8K, original verifier). Runs A and B were repeated at a second seed. A "seed" sets the exam partition, the selection set and the training randomness at once, so a repetition draws a different 150-problem exam set. Most of the later evidence comes from higher-rate runs (Section 4.4; Table S4 in Supplementary Appendix E lists every run). "Five rounds" means round 0 (the untrained model) and four adapters; "three rounds" means round 0 and two adapters, so the round-2 model is a single adapter (about 195 optimizer steps for Run A at 5×10^{-4}) trained on the round-1 pool.

3.4 Models, Data, Training, Metrics

We use Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct (Design P uses both, Design I the 1.5B model only) and the first 500 problems of the GSM8K test split [2] and MATH-500, the 500-problem subset of MATH [5]. The prompt is a single user turn: "Solve the math problem. Put the final answer in `\boxed{}`." followed by the problem. Generation samples at temperature 0.8 with a cap of 512 new tokens; the other decoding settings come from the checkpoint's generation configuration (top-p 0.8, top-k 20, repetition penalty 1.1). Training is LoRA SFT (`transformers` 5.0.0, `peft` 0.19.1, TRL 1.10.0 `SFTTrainer`; rank 16, $\alpha = 32$, on `q_proj` and `v_proj`, LoRA dropout 0, weight decay 0, gradient clipping 1.0 and trainer seed 42, all library defaults read from the pinned sources; batch size 2, no accumulation; one epoch; AdamW with linear learning-rate decay and no warmup; loss over the whole chat sequence, prompt included; fp16 on

one Kaggle T4), with a training sequence limit of 512 tokens that includes the prompt. Because this limit equals the generation cap but also holds the prompt, any kept solution longer than about 512 tokens minus the prompt length is cut in training and loses its final answer and end-of-sequence token (24–26% of the round-0 pool; Section 4.5). All kept solutions are used, so the number of optimizer steps per round is half the pool size. The learning rate is 10^{-5} unless stated otherwise; this value was our code's default and was not tuned. Metrics per round, on the exam set: pass@1 is the correctness of the *first* of the k samples (one draw per problem, so it carries sampling noise); pass@k; *sample accuracy*, the mean over all k samples; style counts over all generations (boxed, dfrac, dollar, ...); and per-problem added/lost counts between consecutive rounds, where a problem is *solved* in a round if at least one of its k samples is accepted. Added minus lost therefore equals $150 \times \Delta$pass@k: the counts show gross churn, not an independent test. pass@1 was fixed in advance; sample accuracy was not, but it estimates the same quantity from all k samples with less variance, so after the runs we chose it as the primary accuracy measure (claims resting on it, including the 2.2-point bound, are post hoc in the choice of measure). "Correct" always means accepted by the verifier (preset `reward`), the same verifier that selects the training data.

Uncertainty. For every within-run change (round 0 against the last round) and every paired comparison of two runs on the same exam partition we report a 95% interval from a paired problem-level bootstrap (10,000 resamples of the 150 exam problems; 2,000 for the matched and audit columns), which respects the clustering of the k samples within a problem. These intervals do *not* cover training-seed variance, which we see only across two or three seeds: the replicate pair we use as a gauge ends 3.2 points of sample accuracy apart (a seed also changes the exam partition, so such pairs mix training and partition variance). We make no multiplicity correction and the runs are not independent (they share exam partitions and round-0 draws), so patterns across runs are descriptive, not tests. Single proportions use Wilson intervals. *Detectable effects.* From the interval half-widths h in Table 2, a change would be detected with 80% power at the 5% level only if it exceeded about $2.8\,h/1.96$: about 10 points of pass@1, 8–10 of pass@k and 3.5–5.5 of sample accuracy on MATH-500. Because h omits training-seed variance (the replicate pair differs by 3.2 points and B minus A flips from +5.3 to −7.3 on near-identical pools, Section 4.4), these are optimistic thresholds, not bounds: even with h as the whole error, a gain at that size is missed 20% of the time, and probably more often (Supplementary Appendix C).

4 Results

4.1 Design P: An Apparent Collapse

In the pooled design the 1.5B model declines severely and with acceleration (pass@1 50.0% → 18.6%, pass@8 72.6% → 48.6%; Table S3, Supplementary

Appendix D), on both datasets (Table S2); the 0.5B decline is small. The accepted pool contracts every round and `\boxed{}` falls from 5,608 to 2,693 of 8,000 generations. This design cannot be read causally: one adapter trains on both pools, it scores on the problems it selects from, and it trains on about 4.5× as many examples per round as the isolated MATH-500 run (3,962 against 872 at round 0) (Supplementary Appendix D).

4.2 Design I at 10^{-5}: Small Declines, a Larger One on GSM8K

On the held-out exam (Fig. S1 and Table S5 in Supplementary Appendix E) pass@8 ends within 4 points of round 0 in every run. On MATH-500, pass@1 is 2.7–4.0 points lower at round 4 (run-log values, see Table 2), with every interval including zero (Table 2); sample accuracy falls 1.2–3.2 points, with an interval excluding zero only for B at seed 0 (A at seed 0 reaches 0.0), and matched finished accuracy 1.1–4.2, with intervals that only just exclude zero for A and B at seed 0. We read this as a possible small decline, not an established one; we see no gain of the size the design could detect. Run C (GSM8K) declines more: pass@1 falls 12.7 points ($[-21.3, -4.0]$), but it took about twice Run A's optimizer steps, so we do not attribute this to the dataset alone. Length and cap-hit share barely change on MATH-500, weak evidence that the adapter did little (Supplementary Appendix E).

4.3 The Patch Control Could Not Detect an Effect

At round 4, B minus A is -0.7 [-3.3, $+2.0$] points of pass@1 at seed 0 and $+0.0$ [-2.7, $+2.7$] at seed 1 (recomputed from the saved generations; Table 2 lists run-log values, -0.7 at seed 1). This null is uninformative: `\dfrac` appears in 0.50–0.73% of generations, so the kept pools differ by at most 4 examples (under 0.5% of 872), and at 10^{-5} the adapter changes the outputs little anyway; the frequent gold-storage defect is in both arms.

4.4 Higher Learning Rates

Because 10^{-5} changed the MATH-500 outputs little, we reran Run A at 10^{-4} (three seeds), 2×10^{-4} (two) and 5×10^{-4} (two, with Run B), with three rounds and $k = 4$ (Table 1). These runs also differ from the 10^{-5} runs in k, rounds, pool size and metric, so only 10^{-4} to 5×10^{-4} isolates the rate. *At* 5×10^{-4} *every A/B run declines:* pass@1 falls 4.0–11.3 points and the sample-accuracy intervals exclude zero in all four A/B runs, though same-seed A and B are closer to two replicates. *The decline may grow with the rate on one seed:* at seed 1 sample accuracy falls 1.5, 3.5 and 8.3 points across the three rates, at seed 2 an almost flat 3.7, 3.8 and 4.7. The cap-hit share rises only at 5×10^{-4}. *The patch has no consistent effect:* B minus A in round-2 pass@1 is +5.3 at seed 1 and -7.3 [-14.0, -0.7] at seed 2, a sign flip from pools differing by 0–3 examples: trajectory noise. *Style does not separate imitation from drift:* boxed share falls

with the pool, which both would produce, and the no-filter and gold arms decline with no verifier in the selection step (they are still scored by it), so generic SFT damage fits as well.

Table 1. Run A (MATH-500, $k = 4$, three rounds) across learning rates and seeds: pass@1, pass@4 and sample accuracy at round 0 $\rightarrow$ round 2, the share of round-2 generations that hit the 512-token cap (re-tokenised to ≥ 508 tokens; round 0 is 43.0–43.2%, 48.2% at seed 3), and problems added/lost over rounds 1–2. The 5×10^{-4} rows are the A rows of Table S6.

Rate	Seed	pass@1	pass@4	Sample acc.	Cap-hit (R2)	Added/lost
10^{-4}	1	33.3 $\rightarrow$ 34.0	50.0 $\rightarrow$ 49.3	33.7 $\rightarrow$ 32.2	42.7%	20/21
10^{-4}	2	31.3 $\rightarrow$ 28.0	51.3 $\rightarrow$ 45.3	30.8 $\rightarrow$ 27.2	42.8%	25/34
10^{-4}	3	29.3 $\rightarrow$ 26.7	40.0 $\rightarrow$ 41.3	28.5 $\rightarrow$ 27.5	45.7%	16/14
2×10^{-4}	1	33.3 $\rightarrow$ 31.3	50.0 $\rightarrow$ 46.7	33.7 $\rightarrow$ 30.2	40.8%	23/28
2×10^{-4}	2	31.3 $\rightarrow$ 25.3	51.3 $\rightarrow$ 45.3	30.8 $\rightarrow$ 27.0	39.7%	21/30
5×10^{-4}	1	33.3 $\rightarrow$ 23.3	50.0 $\rightarrow$ 42.7	33.7 $\rightarrow$ 25.3	46.2%	22/33
5×10^{-4}	2	31.3 $\rightarrow$ 27.3	51.3 $\rightarrow$ 44.7	30.8 $\rightarrow$ 26.2	51.2%	26/36

Two controls at 5×10^{-4}. The *no-filter* run trains on all k samples of every selection problem (1,400 examples per round), removing the verifier; the *train-on-gold* run trains on the 350 reference solutions every round (Table 3). Neither is a single-variable control: no-filter takes 3–4$\times$ Run A's optimizer steps, train-on-gold about the same. Removing the filter is not better than keeping it beyond noise and is lower on pass@4 at both seeds (round-2 differences -2.0 and -5.3 points, intervals including zero), but the dose difference means this cannot be attributed to the filter; the no-filter pool is also truncated far more often in training (62.4% against 24.5% at round 0). Training on reference solutions is followed by a much larger decline (pass@4 down about 21 points at both seeds) while the model adopts the reference style (`$`-delimited spans 5% $\rightarrow$ 98%). Post hoc checks bound verifier error from below only (about 2 points of sample accuracy among boxed answers), so we cannot bound it from above. That no arm, this one included, improves on the untrained model means we have no positive reference: the nulls above may reflect the recipe (LoRA on q/v, loss over the prompt, a 512-token limit) rather than self-training. Run C at 5×10^{-4} (GSM8K, one run) falls 22.0 points of pass@1, from both less accurate and more unfinished solutions (Supplementary Appendix G).

4.5 Anatomy of the Decline

This subsection is post hoc and descriptive. Sample accuracy mixes whether a finished solution is right with whether the model finishes within the 512-token cap. Re-tokenising the saved text, we call a generation *cap-hit* at ≥ 508 tokens. At the rates where the share does not rise, 40–46% of round-2 MATH-500

Table 2. Change from round 0 to the last round (percentage points) with 95% paired problem-level bootstrap intervals over the 150 exam problems (10,000 resamples; `bootstrap_ci.py` in the code repository). Last round is R4 at 10^{-5} ($k = 8$) and R2 elsewhere ($k = 4$). "Finished, matched" is the last-round accuracy of generations below the 512-token cap minus the untrained model's accuracy on the same problem mix (Section 4.5; 2,000 resamples). The intervals do not include training-seed variance. Recomputed values match the reported ones except six round values that differ by one problem (Supplementary Appendix C).

Run, seed	Δpass@1	Δpass@k	Δsample acc.	Finished, matched
Learning rate 10^{-5}				
A, 0	-2.7 $[-9.3, 4.0]$	-1.3 $[-8.0, 5.3]$	-2.4 $[-4.8, 0.0]$	-3.8 $[-7.2, -0.5]$
A, 1	-3.3 $[-10.0, 3.3]$	$+0.0$ $[-6.0, 6.0]$	-1.5 $[-4.1, 1.0]$	-2.3 $[-5.9, 1.2]$
B, 0	-3.3 $[-10.0, 3.3]$	-4.0 $[-10.7, 2.7]$	-3.2 $[-5.9, -0.6]$	-4.2 $[-7.8, -0.9]$
B, 1	-4.0 $[-10.7, 2.7]$	-1.3 $[-7.3, 4.7]$	-1.2 $[-3.7, 1.3]$	-1.1 $[-4.6, 2.6]$
C, 0	-12.7 $[-21.3, -4.0]$	-2.7 $[-7.3, 2.0]$	-13.1 $[-16.7, -9.8]$	-12.8 $[-16.5, -9.4]$
Learning rate 10^{-4}				
A, 1	$+0.7$ $[-6.7, 8.0]$	-0.7 $[-6.0, 4.7]$	-1.5 $[-5.3, 2.2]$	-4.3 $[-9.6, 1.1]$
A, 2	-3.3 $[-9.3, 2.7]$	-6.0 $[-12.7, 0.7]$	-3.7 $[-7.3, -0.2]$	-3.5 $[-9.2, 2.3]$
A, 3	-2.7 $[-10.0, 4.0]$	$+1.3$ $[-4.7, 7.3]$	-1.0 $[-4.3, 2.2]$	-0.6 $[-6.0, 4.7]$
Learning rate 2×10^{-4}				
A, 1	-2.0 $[-8.0, 4.0]$	-3.3 $[-10.0, 2.7]$	-3.5 $[-7.2, 0.0]$	-4.0 $[-8.9, 0.5]$
A, 2	-6.0 $[-12.7, 0.7]$	-6.0 $[-12.7, 0.7]$	-3.8 $[-7.5, -0.2]$	-4.7 $[-10.1, 1.0]$
Learning rate 5×10^{-4}				
A, 1	-10.0 $[-16.7, -3.3]$	-7.3 $[-14.0, 0.0]$	-8.3 $[-12.3, -4.5]$	-9.1 $[-14.2, -4.2]$
A, 2	-4.0 $[-11.3, 3.3]$	-6.7 $[-13.3, 0.0]$	-4.7 $[-8.2, -1.2]$	-5.5 $[-11.7, 0.1]$
B, 1	-4.7 $[-12.0, 2.7]$	-2.0 $[-8.7, 4.7]$	-6.7 $[-10.5, -2.8]$	-6.8 $[-12.4, -1.0]$
B, 2	-11.3 $[-17.3, -5.3]$	-11.3 $[-18.0, -4.7]$	-9.2 $[-12.8, -5.5]$	-5.9 $[-10.9, -1.0]$
No filter, 1	-6.0 $[-12.7, 1.3]$	-9.3 $[-16.7, -2.7]$	-8.2 $[-12.2, -4.3]$	-10.6 $[-16.4, -5.5]$
No filter, 2	-8.7 $[-16.0, -1.3]$	-12.0 $[-19.3, -4.7]$	-10.2 $[-14.2, -6.3]$	-6.6 $[-11.4, -2.0]$
Gold, 1	-18.0 $[-26.7, -10.0]$	-20.7 $[-28.7, -12.7]$	-19.7 $[-25.3, -14.2]$	-25.6 $[-32.1, -18.9]$
Gold, 2	-21.3 $[-28.7, -14.0]$	-20.7 $[-29.3, -12.7]$	-18.7 $[-23.7, -13.7]$	-25.9 $[-31.7, -20.0]$
C, 1	-22.0 $[-29.3, -15.3]$	-9.3 $[-15.3, -3.3]$	-15.3 $[-19.8, -11.2]$	-12.1 $[-16.6, -7.9]$

Table 3. Controls at 5×10^{-4} (MATH-500, $k = 4$, three rounds), on the exam partition of Run A at the same seed. Pool is the kept pool at round 0 $\rightarrow$ round 2; the round-2 model trains on the round-1 pool (392 and 386 examples, about 195 steps).

Run	Pool	pass@1			pass@4		
		R0	R1	R2	R0	R1	R2
Seed 1							
A (verifier-filtered)	440→326	33.3	32.7	23.3	50.0	49.3	42.7
No filter	1400	33.3	26.7	27.3	50.0	49.3	40.7
Train on gold	350	33.3	18.0	15.3	50.0	33.3	29.3
Seed 2							
A (verifier-filtered)	424→350	31.3	31.3	27.3	51.3	46.0	44.7
No filter	1400	31.3	34.7	22.7	51.3	48.7	39.3
Train on gold	350	31.3	8.7	10.0	51.3	28.7	30.7

generations already hit the cap (Table 1), so the cap binds from the start. At 5×10^{-4}, compared with the untrained model on the same problem mix, finished

generations lose 5.5–10.6 points, with intervals excluding zero in five of six runs (the six include two same-seed A/B pairs, which are close to replicates), while the cap-hit share rises 2–14 points. The matched measure conditions on which problems finish after training, so it is descriptive. A three-part decomposition attributes 12–30% of the decline to the cap share in the three original seed-1 runs (and −2 and 5% in two further runs) and 52–72% at seed 2 (Table S9, Supplementary Appendix H). At 10^{-4} and 2×10^{-4} the cap-hit share does not rise and finished accuracy carries the whole change. Train-on-gold is different: responses shorten by more than half and finished accuracy collapses. Retraining without truncated examples, or re-scoring at a 1,536-token cap, leaves the deficit in place (Supplementary Appendix H).

4.6 A Final Round with a Revised Recipe

Because no earlier arm improved, we ran one last round with several recipe changes made at once, aimed at the suspected causes (no ablation of any single one): loss on the completion only, LoRA on all linear layers, a 1,024-token cap and training limit, normalised gold answers, and a pool of 300 problems from the MATH training split, so that the exam is all 500 MATH-500 problems. It is 10^{-4} with $k = 2$ and a single adapter (round 0 to round 1); the loop has three seeds, the gate one. The pool comes from the PRM800K training split, which is the complement of MATH-500 within MATH, so it is disjoint by construction (no content-level overlap check). The verifier-filtered loop (298 examples kept) is compared with training on the 300 reference solutions, a positive gate that must raise accuracy for the recipe to count as able to help.

Table 4. Final round, MATH-500 (500 exam problems, $k = 2$, 10^{-4}). Change from each run's own untrained round 0 to round 1 in points (round-0 pass@1 is 49.2, 46.8, 48.2 at seeds 1–3, from sampling), with 95% paired problem-level bootstrap intervals that omit training-seed variance. The gold arm has one seed.

Arm	pass@1	pass@2	Sample acc.
Untrained (seed 1)	49.2	59.4	–
Loop, seed 1	−4.8 [−9.0, −0.6]	−3.0 [−6.8, +0.6]	−2.9 [−5.8, 0.0]
Loop, seed 2	−3.2 [−6.8, +0.6]	−6.2 [−9.8, −2.4]	−5.3 [−8.1, −2.4]
Loop, seed 3	−4.2 [−8.4, −0.2]	−5.6 [−9.4, −1.6]	−5.4 [−8.3, −2.4]
Loop, mean of seeds	−4.1	−4.9	−4.5
Train on gold	−32.2 [−36.8, −27.6]	−32.0 [−36.6, −27.2]	−30.2 [−33.8, −26.6]

The gate fails. Training on reference solutions lowers pass@1 from 49.2% to 17.0%, and the model adopts the reference style ($-delimited spans in 60 of 1,600 generations at round 0, 1,577 at round 1, counted over the exam and selection problems), as in the earlier train-on-gold arms. A rerun at 2×10^{-5} fails the same way (49.2% → 20.6%, $-spans in 1,558 of 1,600 generations), so the failure is not specific to the rate. It is not a format defect: on the 500 exam problems

the share of generations with a `\boxed` answer is unchanged (90.6 → 90.3%), accuracy among them falls from 48.5 to 17.1%, and the median length drops 3.2× (1,332 → 414 characters; string-match grader; `gate_anatomy.py`). The model imitates terse reference solutions and loses reasoning steps; the three loop seeds show a different pattern (median length 1,332–1,348 → 1,454–1,491 characters, boxed share −0.6, −0.6, −5.1 points, accuracy among boxed answers −2.5, −5.7, −3.4), so their decline is not this effect. so reference solutions are a weak positive control for an instruction-tuned model. The gate therefore fails to validate the revised recipe. With no recipe known to raise accuracy, the loop's decline cannot be attributed to self-training or to the verifier. The verifier-filtered loop loses pass@1 in all three seeds (4.8, 3.2, 4.2 points; two intervals exclude zero) and sample accuracy in all three (intervals exclude zero for seeds 2 and 3). The mean pass@1 change over the three seeds is −4.1 (t interval $[-6.1, -2.1]$), and −4.5 for sample accuracy ($[-8.1, -1.0]$). The loop does not help; the estimates are negative in all three seeds, but with three seeds, one exam draw and one rate we do not claim that it costs accuracy or isolate the cause. The detectable effect is about 6 points of pass@1 ($2.8h/1.96$), against about 10 before.

Injected bias. As intended positive controls for the audit (neither turns out to be one) we reran the loop once each (seed 1, 10^{-4}) with a selection verifier that rejects `\boxed` answers (exam verifier clean). In the first, a hash-chosen 30% of problems are affected: the `\boxed` share falls 4.9 points on them and 3.1 on the others (difference −1.8, $[-7.4, 3.8]$); a hash cannot be learned from the problem, so this control could not elicit selective avoidance. In the second the rule is visible: every `\boxed` answer is rejected when the gold contains `\frac` (49 of the 300 selection problems, 263 examples kept instead of 298; 70 of the 500 exam problems, where the shares are measured). The share on those problems stays at 95.0% and on the others goes from 90.2% to 87.4% (difference +2.8, $[-2.8, +8.3]$), against −3.7, −2.7 and −5.5 in the three unbiased seeds; pass@1 falls 7.0 points ($[-11.2, -3.0]$), more than in the unbiased seeds, which we do not interpret (one run). In neither run does the `\boxed` share fall selectively where boxing is penalised, so neither validates the audit.

5 Discussion

5.1 Answer to the Research Question

Does the model get better at mathematics? Not detectably. No run raises pass@1 or pass@k with an interval excluding zero, at any rate; in the 500-problem final round pass@1 falls 3.2–4.8 points in each of three seeds. The sample-accuracy figure of about 2.2 points is the largest upper limit among observed intervals, a post hoc figure that omits seed variance, not a power statement; on the pre-set measures the design could detect only about 10 points of pass@1 and 8–10 of pass@k (6 and 5 in the final round). Because no arm improves and the positive gate failed, the null belongs to this recipe, and the study cannot say whether self-training in general helps. Every score except the final-round check below is graded by the verifier that selects the data (re-scoring five runs with

V' leaves the decline, Section 3). On the final-round generations a second grader (normalised string match of the last `\boxed` against the gold, no `math-verify`; `indep_grade.py`) gives pass@1 changes of -3.6, -2.2, -4.6 for the three seeds (-30.0 for the gold arm) and sample-accuracy changes of -2.5, -5.5, -5.3 (same sign and size). It reads only the last `\boxed`, so a falling boxed share lowers both graders; it does not rule out a format artefact in the gate, and earlier runs are not independently graded.

Does it learn to write what the verifier accepts? Undecided: the pre-set pattern requires pass@1 to rise, which never happens detectably (largest rise $+0.7$ $[-6.7, +8.0]$), no run varies the frequent gold-storage defect, and the audit of Section 5.3 is too weak.

5.2 Interpretation

One explanation of the Design P curves is a self-reinforcing shrinking pool shaped by the biased verifier. The data neither support nor rule it out: the patch tests a rare bias, and the no-filter control declines without a selection verifier but with more dose. *Removing a rare format defect, or the filter entirely, made no consistent difference*; neither tests the systematic false positives Egashira et al. find harmful in RLVR.

5.3 A Proposed Audit for GVT with a Biased Verifier

Our pre-set rule could not speak once accuracy fell. From post hoc analyses we propose four checks for self-training with a rule-based verifier (untested protocol).

(i) A per-round false-reject audit. Fix a repaired verifier V' that corrects a known defect class of the training verifier V. At each round t, on the loop's own generations, compute

$$r_t = \frac{\#\{\text{accepted by } V' \text{ but rejected by } V\}}{\#\{\text{accepted by } V'\}}.$$

If the model learns to write what V accepts, r_t may fall over rounds, but it also falls when fewer answers are boxed: since V' differs from V only on boxed predictions, $r_t = r_t^{\text{boxed}} \times$ (boxed share among V'-accepted) holds as an identity. This is the per-step tracking of RLVR studies [4,16] applied to offline SFT (Table 5; `rt_audit.py`). For the gold-storage defect it falls in all three self-training runs we re-scored, by 2.6, 2.6 and 1.5 points, with intervals reaching or just touching zero. The fall is accounted for by the falling `\boxed` share among correct answers ($91 \to 76\%$ and $93 \to 78\%$ at 5×10^{-4}): an identity, not a test, so adaptation is not excluded. On boxed answers alone the changes (-0.4 to $+0.7$) have intervals that contain the overall fall of the same run, so "flat" means not detected. The defect is bound to the problem: it fires on 54–62 problems, rejecting 94–98% of correct answers, and 29–34 of them have an unparseable gold. On the 24–28 affected problems where some form could, r_t moves by -5.5, -1.6 and -0.2

points (intervals 10–26 wide), and the pool holds only 2–12 such examples per round, so we expect weak pressure to adapt (untested). The two injected-bias runs (Section 4.6) show no selective avoidance, but cannot tell us whether the audit would detect it. Train-on-gold, with no verifier in the loop, gives r_t rises of 2.1–3.3 points (Table 5), so r_t drifts as much as the falls we observe, unrelated to the verifier. It should be reported on the answers and problems where V and V' can differ, with the count of training examples that could teach avoidance.

Table 5. The false-reject audit for the gold-storage defect (post hoc), in percent (all 500 problems, selection problems included), round 0 → last round, with the change and a 95% problem-level bootstrap interval (2,000 resamples). "Boxed only" restricts to predictions containing `\boxed`; "affected, parseable" to problems where the defect ever fires and V can parse the gold. Pool: examples per round from affected problems that V accepts.

Run	r_t subset	R0	Last	Δ	95% interval	Pool
A, 5×10^{-4}, seed 1	all	18.3	15.7	−2.6	$[-5.3, 0.3]$	3–5
	boxed only	20.0	20.7	+0.7	$[-2.9, 4.3]$	
	affected, parseable	92.8	87.2	−5.5	$[-20.1, 6.3]$	
A, 5×10^{-4}, seed 2	all	18.1	15.5	−2.6	$[-5.2, 0.1]$	2–3
	boxed only	19.4	19.8	+0.4	$[-2.8, 3.4]$	
	affected, parseable	95.1	93.5	−1.6	$[-14.7, 9.5]$	
A, 10^{-5}, seed 0	all	18.3	16.8	−1.5	$[-2.9, 0.0]$	9–12
	boxed only	19.9	19.5	−0.4	$[-2.0, 1.2]$	
	affected, parseable	92.3	92.0	−0.2	$[-4.9, 4.6]$	
Gold, seed 1	all	18.2	21.5	+3.3	$[-1.2, 8.1]$	—
	boxed only	20.0	22.8	+2.8	$[-2.0, 7.7]$	
	affected, parseable	95.5	94.7	−0.7	$[-9.1, 6.2]$	
Gold, seed 2	all	18.0	20.2	+2.1	$[-2.2, 6.7]$	—
	boxed only	19.4	21.1	+1.7	$[-3.3, 6.5]$	
	affected, parseable	95.1	97.6	+2.5	$[-4.8, 11.9]$	

(ii) Decompose declines under a fixed cap. Report the change in sample accuracy split into a finished-accuracy part and a cap-share part (Table S9, Supplementary Appendix H). The split is descriptive: at 5×10^{-4} the cap share accounts for 12–30% of the decline in the three seed-1 runs and 52–72% at seed 2.

(iii) Check the dose of a control first (our `\dfrac` patch changed 0–4 examples) and *(iv) match dose across designs.*

6 Limitations

Unequal training doses confound every cross-arm comparison, and the controls are underpowered: the patch changes 0–4 training examples, and the no-filter and train-on-gold controls have two seeds each. *Intervals cover only problem draw and sampling,* not training-seed variance; the final-round loop has three

seeds, the gate and the bias controls one, and there is no multiplicity correction. *The learning-rate grid is coarse* and changes k, rounds and dose with the rate. *Power is limited* (about 10 points of pass@1 with 150 exam problems, about 6 in Section 4.6), and *the adaptation audit is weak* (24–28 problems, no valid positive control). *The verifier audit* used other models' solutions and inspected only rejections; the gold-storage defect ran unvaried through every earlier run, whose headline numbers are graded by the biased verifier that also selects the data, and the final-round string-match grader shares the `\boxed` dependence. The final-round decline and the gate failure cannot be attributed to one recipe change. *Truncation:* the 512-token cap binds from the start (a quarter of the pool is cut). *Scope:* one model family and size, all instruction-tuned (no base model), one verifier, a restart-from-base design that leaves compounding little room (no continued training), unquantified Qwen2.5 contamination on MATH/GSM8K, and a default repetition penalty (1.1).

7 Conclusion

The design detected no pass@1 or pass@k gain at any rate (it could see only gains of about 10 and 8–10 points; 6 in a final 500-problem round, where pass@1 falls 3.2–4.8 points in all three seeds). Because training on reference solutions lowers pass@1 by about 30 points even with a revised recipe, this is an inconclusive null for the recipe and says little about self-training; since scores are graded by the selecting verifier (apart from one string-match check), neither branch of the question is answered. The post hoc false-reject audit is uninformative about adaptation: its fall is tied to the falling `\boxed` share by an identity. The pooled decline is absent in the isolated runs (4.5× fewer examples): dose and pooling are confounded.

Next come a dose-matched control for the gold-storage defect, a recipe that passes the gate, and a second grader (Supplementary Appendix K).

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Code availability Code, saved generations and analysis scripts are at https://github.com/tsan2711/DACNTT_2026 (tag `paper-2026-11b`). Generation-time token counts, Design P's training logs and the 1,536-token re-scoring outputs were not saved.

Use of AI tools The authors used Claude (Anthropic) to help write the manuscript and the analysis and training scripts, and to run automated reviews. The authors designed the study, ran the experiments and take responsibility for the content.

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